MILITARY SPECIFICATION

LIGHT, IGNITION TIMING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a solid state, high intensity, ignition timing light that derives its power from vehicle electrical ignition systems while testing those commercial and military ignition systems.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

J-C-580	-	Cord, Flexible, and Wire Fixture (Electrical, 0- to 600-Volt Service).
W-C-440	-	Clips, Electrical, General Specification For.
QQ-S-571	-	Solder, Tin Alloy: Tin-Lead Alloy; and Lead Alloy.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-C-843	-	Cushioning Material, Cellulosic.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-EST-S (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4910

AMSC N/A

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MIL-P-116 - Preservation, Methods of.
MIL-T-152 - Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-C-3702 - Cable, Power, Electrical: Ignition High-Tension.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods.
MIL-STD-1460 - Procedure for Soldering of Electrical Connections and Printed Wiring Assemblies.
MS3368 - Strap, Tiedown, Electrical Components, Identification, Adjustable, Self-Clinching, Plastic, Type II, Class 1.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These

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documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

3.2 Design. The timing light shall be solid-state, and designed to measure ignition spark timing of internal combustion engines for military or commercial vehicles that use battery distributor ignition systems. The timing light (flash tube) shall be triggered by indirect inductive pick-up of the vehicle's secondary ignition voltage pulses (see 3.5.1). The timing light shall meet the requirements of this specification when used on military vehicles having shielded type ignition systems with igniter type distributors, including those with built-in ballast resistors, and such military vehicles as the following:

Truck, Cargo, 1 1/4 Ton, 4X4, M880
Truck, Cargo, 1 1/4 Ton, 4X2, M890

The timing light shall be either a one-piece gun type with a carrying case, or a two-piece unit consisting of a flash tube assembly and a power pack contained within a carrying case (see 3.6 and 3.6.5). The timing light shall also meet the requirements of this specification when used on commercial type ignition systems, including all solid-state, breakerless, High Energy Ignition (HEI) type systems. The timing light shall be designed to operate from a direct current (dc) power source of 12 volts while checking regulated 12 volt ignition systems, negative ground. The timing light shall produce high intensity light flashes and shall operate in any position. Those parts of the light subject to wear, breakage, or distortion shall be accessible for adjustment, repair, and replacement without modification, distortion, or damage to the light or any component.

3.2.1 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the timing light. In this specification, all measurements, dimensions, sizes, and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in FED-STD-376.

3.2.2 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.2.3 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

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3.2.4 Safety. The timing light shall provide fail-safe (see 6.4.1) features for safety of personnel during the installation, operation, maintenance and repair, or interchanging of a complete assembly or components parts thereof. The timing light shall, in addition, comply with the applicable requirements of MIL-STD-454. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1).

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.4 Construction. The timing light shall be furnished complete so that it can be used for the operations specified herein. Parts used in fabricating the light shall be new and free of defects. All part surfaces shall be clean and free of extraneous materials. External surfaces shall be smooth and free of sharp edges. Functional surfaces shall be free from burrs and defects affecting performance.

3.4.1 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for adjustment and for preventing loss of tightness. The methods for adjustment and for preventing loss of tightness shall be in accordance with accepted engineering standards and practices. All such parts when subject to renewal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

3.4.2 Electrical connections. Connections of conductors and terminal parts shall be of the screw, pressure, or solder type. When soldered connections are used, the connections shall be mechanically secured before soldering. Solid-state applications, where connections cannot be mechanically secured, shall be soldered to maintain proper electrical continuity. Only rosin base fluxes shall be used in soldering operations. Connections to screw-type terminals shall be mechanically secured to prevent loss of tightness. The solder alloy shall conform to composition Sn60, Sn62 or Sn63 of QQ-S-571. Soldering shall comply with the applicable requirements of MIL-STD-1460.

3.5 Performance and product characteristics.

3.5.1 Performance characteristics. The timing light shall flash at rates of 3.75 through 25 flashes per second corresponding to engine speeds of 450 through 3,000 revolutions per minute (rpm) without spurious flashing. The timing light (flash tube) shall be triggered indirectly (see 3.2 and 3.6.4.2) by secondary ignition voltage pulses from 1.2 kV to 40 kV, zero to peak, and shall not be damaged by potentials up to 46 kV, zero to peak, measured at a frequency of 10 pulses per second for 30 seconds without arcing or flashover. The timing light, when powered by 14.5 volts dc $\pm$0.3 volts and flashing at the rate of 20 flashes per second, shall provide a minimum illumination of ten lumens per square foot (lpsf) at a distance of 12 inches and 6.5 lpsf at a distance of 20 inches when measured from the face of the lens to the target with a Weston (or approved

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equal) light meter, Model 756 on the 0-20 scale range, equipped with Viscor and cosine correction filters. The duration of flash shall be no less than eight microseconds and no more than 15 microseconds. Engine operation shall not be impaired by operation of the timing light. When used for timing ignition systems, the light shall not double flash (due to extreme sensitivity) or miss. The timing light shall not be damaged by any reverse polarity conditions. It shall be capable of operation throughout a temperature range of 0° Fahrenheit (F) to +115°F (see 3.5.6.1).

3.5.2 Endurance. The timing light shall withstand, without damage, continuous use for 16 hours when operated from an input power source of 12 to 16 volts dc. The timing light shall be capable of being operated eight hours at 12 volts and eight hours at 16 volts.

3.5.3 Vibration. The timing light in its carrying case with the cover secured, shall show no damage after being subjected to a vibration (simple harmonic motion), having an amplitude of 0.03 inch (0.06 inch maximum total excursion) through a frequency range of 5 to 55 hertz (Hz) and return to 5 Hz, executed once per minute for a period of one hour through each of its three major axes.

3.5.4 Shock requirements. The timing light, enclosed in its carrying case with the cover latched, shall not be damaged and shall meet the performance requirements of 3.5.1 and 3.6.3.2 after being dropped at least six times from a height of at least four feet to a bare concrete floor on each of three different axes.

3.5.5 Dielectric strength. The timing light shall show no evidence of smoking, arcing, flashover, charred insulation, or other damage when subjected to the application of 500 volts root mean square (rms), 60 Hz, for a period of one minute between mutually insulated portions and between insulated portions and ground.

3.5.5.1 High Secondary Voltage. The timing light shall operate as specified in 3.5.1 and shall withstand, without double flashing, smoking, arcing, flashover, charred insulation or other damage, 46 kV, zero to peak, measured at a frequency of 10 pulses per second and applied for 30 seconds.

3.5.6 Temperature and humidity conditions.

3.5.6.1 Operating temperatures. The timing light shall be capable of meeting the performance requirements of 3.5.1 while being subjected to an ambient temperature range between 0° ±5°F and +115°F ±5°F.

3.5.6.2 Humidity conditions. The timing light shall withstand, without damage, and shall meet the requirements of 3.5.1:

(a) After having been exposed for 20 hours to, and while at, a relative humidity of 93 to 97 percent in an air temperature of +73° to +77°F, air movement and radiation negligible.
(b) After having been exposed for four hours to, and while at, a relative humidity of 5 to 10 percent in an air temperature of +110° to +115°F, air movement and radiation negligible.

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3.6 Components. The timing light shall consist essentially of a flash tube, power pack, lens, and housing enclosed in a carrying case. These components may be furnished as a one-piece (gun type) unit, or as a two section unit consisting of a flash tube assembly connected to the power pack with a flexible cable at least 60 inches long. Both the one-piece unit and the two section unit shall be enclosed in a carrying case (see 3.6.5). On the one-piece (gun type) unit, the housing shall be not more than 12 inches long over-all with the barrel not more than two inches in circumference at any point, excluding the handle. The handle shall be not less than four nor more than five inches long measured from the barrel and shall be five to seven inches in circumference. The design of the timing light shall be such that spark plug voltages shall not cause insulation punctures on any portion of the unit. The assembly shall be constructed so that flash tubes can be replaced under field conditions.

3.6.1 Flash tube. The flash tube shall be a polarized, plug-in type. It shall not emit radiation above 600 nanometers.

3.6.2 Lens. The lens may be separate from the flash tube or may be incorporated in the flash tube. The lens shall be recessed from the housing edge for protection against damage.

3.6.3 Power pack. The power pack shall provide the correct operational voltage to the flash tube. There shall be no internal batteries used. All components shall be mounted to be removable as a complete unit. A voltage divider circuit shall not be used to correct voltage transformers. Electromagnetic vibrators shall not be used in any portion of the timing light.

3.6.3.1 Transformers. Transformers shall be capable of meeting the performance requirements of this specification, and shall not exceed a temperature rise of 70°F over ambient temperature during eight hours of continuous operation.

3.6.3.2 Over-voltage protection. A relay or solid-state circuit protection device shall be incorporated in the circuitry to prevent damage to the power pack, and flash tube assembly in case voltages over 16 and up to 50 volts are encountered.

3.6.3.3 Relays. All relays used in the power pack shall be enclosed but, not necessarily hermetically sealed.

3.6.4 Leads. Leads shall be anchored inside the case and shall be protected against strain at electrical connections by means of clamps. Leads passing through metal enclosures shall be protected by rubber grommets. Leads shall be mechanically secured. If a two section unit is furnished, the cable connecting the flash tube assembly to the power pack shall have a minimum length of 60 inches and shall be a heavy-duty, flexible, stranded, No. 18 American Wire Gage (AWG), type HPN cord conforming to J-C-580.

3.6.4.1 Battery leads. Battery leads shall be a minimum of 120 inches in length, and shall be heavy-duty, flexible, stranded, No. 16 AWG, type HPN cord conforming to J-C-580. The free ends of the battery leads shall be terminated with insulated steel clips, type PC, 25 ampere capacity conforming to W-C-440, with red (positive) and black (negative) insulators.

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3.6.4.2 Spark plug lead. The timing light shall be provided with a spark plug lead a minimum of 60 inches in length and shall be in accordance with designation M3702-S7LC of MIL-C-3702. The lead shall be designed to allow for indirect inductive pick-up of the vehicle secondary ignition impulses (see 6.4.2). The indirect inductive pick-up shall be of the fully insulated spring clamp-on or clamp-around (spark plug cable) type. For information, the military spark plug adapter ignition lead for access to shielded ignition systems is high tension ignition cable, type U7LC, of MIL-C-3702, with approximately two inches of cable available for access by an indirect inductive pick-up.

3.6.5 Carrying case. Both the one-piece and the two section timing light shall be contained in a metal carrying case with a hinged cover. The carrying case shall be designed and constructed to prevent damage to any component (case or timing light) when exposed to the vibration and shock requirements of 3.5.3 and 3.5.4. The case and cover shall be fabricated of sheet metal not less than Manufacturers' Standard Gage No. 20 (0.036 inch). Compartments holding leads shall be sized so that no special coiling arrangement is required for lead stowage. Operating instructions and caution notes shall be legibly and permanently marked either on top of the power pack unit or on the inside surface of the cover. The hinged cover shall be held in a closed position by one or more tension fasteners. The case shall have a carrying handle which allows a comfortable four finger grip. Oil-resisting synthetic rest buttons shall be attached to the case near each bottom corner and near each corner on the side opposite the carrying handle. Overall dimensions of the complete case, including handle, fasteners, and rest buttons, shall not exceed 9 by 10 by 5 inches for the two section unit. The carrying case for the one-piece unit shall be of sufficient size to house the complete timing light. All exterior corners shall be rounded.

3.7 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.8 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.9 Identification of product. The timing light shall be marked for identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number. On the one-piece unit, both the timing light and the carrying case cover exterior shall be marked with the product identification. On the two section unit, the carrying case cover exterior shall be marked with the product identification.

3.9.1 Marking. The timing light shall be permanently marked to indicate 12 volt battery operating power and indirect inductive pick-up from the secondary ignition impulse (distributor to spark plug lead). In addition, identification tags conforming to MS3368-5 shall be affixed to the battery leads and permanently marked with +12V for the positive battery lead and -12V for the negative battery lead.

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3.10 Workmanship. Standards of workmanship shall assure that the timing light shall have the stability, strength, durability, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. When required (see 6.2.1), first article inspection shall be applied to a preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production items offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (e) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Performance test (see 4.6.1).
(c) Voltage protection test (see 4.6.2).
(d) Trigger voltage test (see 4.6.5).
(e) Packaging inspection (see 4.7).

4.4.1 Sampling. Unless otherwise specified (see 6.2.1), sampling for quality conformance inspection shall be in accordance with MIL-STD-105, using the following Inspection Levels and Acceptable Quality Levels (AQL):

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Inspection or Test	Inspection Level	AQL
Product examination	100 percent	-
Performance test	II	1.0
Voltage protection test	II	1.0
Trigger voltage test	II	1.0
Packaging inspection	S-4	4.0

4.5 Product examination. Visually and manually examine the timing light to determine conformance with the requirements of 3.2 and 3.3, 3.4 through 3.4.2, and 3.6 through 3.10. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the timing light has passed the examinations prescribed for it by the contractor's inspection system shall be cause for rejection.

4.6 Tests.

4.6.1 Performance test. A test circuit mock-up which simulates the ignition system of vehicles listed in 3.2, and the inductive pick-up restrictions of 3.6.4.2, shall be used to determine compliance with 3.5.1. The timing light shall be tested for a period of one minute at each of the following engine rpm: 450, 600, 1,200, 1,800, 2,400 and 3,000. The duration of the flash tube current, as measured by an oscilloscope having a minimum sweep rate of 100 nanoseconds (or less) per centimeter shall be no less than eight and no more than 15 microseconds. While flashing at the rate of 20 flashes per second, the light shall provide a minimum illumination of 10 lpsf at a distance of 12 inches and 6.5 lpsf at a distahce of 20 inches, when measured on the 0 to 20 fc scale of a Model 756 Weston light meter (or approved equivalent) equipped with Viscor and cosine correction filters. All measuring instruments shall be accurate to within one percent of the value being read. At least once during this test the battery leads shall be reversed to verify compliance with 3.5.1. This test shall be conducted twice, once on a simulated 12 volt and once on a 16 volt ignition system. Failure of the timing light to flash within the duration requirement at all listed engine speeds and voltages, failure of the light to meet the illumination requirements, or failure of the light to withstand the battery lead reversals shall be cause for rejection.

4.6.2 Input voltage and over-voltage protection test. The timing light shall be operated over a range of 12 to 16 volts dc, at any one of the rpm values in 4.6.1, to verify proper operation of the flash tube (no misses or double flashes). The input voltage shall be increased from 16 to 50 volts dc. Damage to the power pack or flash tube assembly shall be cause for rejection. Following the over-voltage test, the timing light shall be operated at both 12 and 16 volts dc. The flash tube shall not miss or double flash.

4.6.3 Endurance test. Using the ignition system mock-up, the timing light shall be operated continuously at +70° ±10°F in accordance with the following conditions:

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Test run	Timing light operating voltage	Spark gap	Flash rate per second	Time per test run
1	12 vdc	0.030	25	8 hrs.
2	16 vdc	0.030	25	8 hrs.

The temperature rise of the transformer shall not exceed 70°F (see 3.6.3.1). At the conclusion of the endurance test the timing light shall be checked for performance in accordance with 4.6.1 and 4.6.2.

4.6.4 High secondary voltage test. If the direct secondary voltage pick-up is used to trigger the flash tube (not an inductive pick-up), the high voltage circuit shall be subjected to the following test. A pulsed triggering voltage potential of 46 kV, zero to peak, at a pulsed rate of 10 pulses per second shall be applied to the pick-up lead for 30 seconds. At the conclusion of this test, the timing light shall be inspected for evidence of damage. Evidence of smoking, arcing, flashover, charred insulation or other damage shall be cause for rejection. The timing light shall then be tested in accordance with 4.6.1 and 4.6.2.

4.6.5 Trigger voltage test. The triggering device of the flash tube shall be tested, using the circuit mock-up (see 4.6.1), at a regulated input voltage of 14.5 volts dc ±0.3 volts at simulated engine speeds of 450 and 3000 rpm. An input trigger voltage of 1.2 kV, zero to peak, shall be applied to the trigger at both engine speeds. An input trigger voltage of 40 kV, zero to peak, shall be applied to the trigger for both engine speeds. Any miss-flash or double flashing shall be cause for rejection (see 3.5.1).

4.6.6 Dielectric strength test. The timing light shall be subjected to 500 volts rms at 60 Hz between each battery lead clip and all exposed metal parts of the timing light housing and power pack housing for a period of one minute for each battery lead. Following the test, the complete timing light shall be inspected for damage and subjected to the tests in 4.6.1 and 4.6.2. Failure of these tests shall be cause for rejection.

4.6.7 Operating temperature tests. The complete timing light shall be subjected to the low temperature condition of 3.5.6.1 until stabilized. While at that temperature, the timing light shall be tested in accordance with 4.6.1 and 4.6.2. The test shall then be repeated for the high temperature condition of 3.5.6.1. Failure of the light to pass the tests of 4.6.1 and 4.6.2 shall be cause for rejection.

4.6.8 Humidity test. The complete timing light shall be exposed to the two environmental conditions specified below (see 3.5.6.2). Immediately after each exposure, the light shall be tested as specified in 4.6.1 and 4.6.2.

(a) For 20 hours at an air temperature of +73° to +77°F, at a relative humidity of 93 to 97 percent, air movement and radiation negligible.
(b) For four hours at an air temperature of +110° to +115°F at a relative humidity of 5 to 10 percent, air movement and radiation negligible.

4.6.9 Vibration test. The complete timing light shall be secured in its carrying case and subjected to the vibration specified in 3.5.3. The timing light

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submitted to the Government for acceptance comply with all requirements of the

shall then be tested in accordance with 4.6.1 and 4.6.2. Failure of the light to pass the tests, or permanent deformation or breakage of any component or the case, shall be cause for rejection.

4.6.10 Shock. The timing light, secured in its carrying case with the cover closed, shall be dropped six times a vertical distance of four feet to a horizontal bare concrete surface on each of three mutually perpendicular axes. Following the test, the complete light shall be subjected to the tests of 4.6.1 and 4.6.2. Failure of the light to pass the tests, or the fracture or breakage of any component of the timing light or the case, shall be cause for rejection.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Industrial. Unless otherwise specified (see 6.2.1), Industrial preservation shall be provided. When military (Level A) preservation is required, it shall be so specified in the contract.

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The timing light shall be cleaned in accordance with process C-1 and dried in accordance with procedures D-1 or D-4 of MIL-P-116.

5.1.1.2 Unit packaging. Each timing light shall be packaged in accordance with method IC-2 of MIL-P-116, using a fiberboard box conforming to PPP-B-636, class domestic, as the unit container. The leads shall be coiled to a minimum safe diameter and secured to prevent uncoiling. Technical data issued with the timing light shall be placed in a plastic envelope. Closure of the envelope shall be by heatsealing or stapling. Components and technical data shall be placed in their respective stowage compartments of the carrying case. Adequate cushioning conforming to PPP-C-843, type II, shall be applied to restrict movement of the items.

5.1.2 Industrial. Industrial preservation shall be as specified in ASTM D3951-82.

5.2 Packing. Packing shall be Level A, Level B, or Industrial. Unless otherwise specified (see 6.2.1), Industrial packing shall be provided. When military (Levels A and B) packing is required, it shall be so specified in the contract.

5.2.1 Level A. Timing lights packaged as specified in 5.1.1.2 shall be packed in an exterior shipping container conforming to PPP-B-601, overseas type, or PPP-B-621, class 2. The gross weight of each shipping container shall not exceed the weight limitations of the applicable box specification. As far as practicable, containers shall be uniform in size and contain like quantities. Blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186. Container closure and strapping requirements shall be in accordance with the appendix to the applicable box specification.

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5.2.2 Level B. The timing lights shall be packed as specified in 5.2.1 except the shipping container shall conform to PPP-B-640, class 2.

5.2.3 Industrial. Industrial packing shall be as specified in ASTM D3951-82.

5.3 Marking. Marking shall be Military or Industrial. Unless otherwise specified (see 6.2.1), Industrial marking shall be provided. When military marking is required, it shall be so specified in the contract.

5.3.1 Military marking. Military marking shall be as specified in MIL-STD-129.

5.3.2 Industrial marking. Industrial marking shall be as specified in ASTM D3951-82.

6. NOTES

6.1 Intended use. The timing lights are intended for use in checking and testing the timing and spark advance of internal combustion engines with 12 volt battery-distributor ignition systems.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Exceptions and additional safety/health requirements when required (see 3.2.4).
(d) Painting and finishing if different (see 3.8).
(e) Product identification if different (see 3.9).
(f) First article inspection when required (see 4.3).
(g) Sampling if different (see 4.4.1).
(h) Level of preservation, packing, and marking if different (see 5.1, 5.2, and 5.3).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.4 Definitions.

6.4.1 Fail-safe. The design feature of a part, unit, or equipment which causes the item to fail only into a non-hazardous mode.

6.4.2 Secondary ignition impulse. Secondary ignition impulse is the peak electrical energy in the secondary side of the ignition system, which travels from the coil through the distributor to spark plugs by ignition cables, sufficient to fire the air and fuel mixture in the cylinder upon the compression stroke of an internal combustion engine.

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6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - YD
Air Force - 99

Review activities:
Navy - AS, MC
Air Force - 82
DLA - CS

Preparing activity:

Army - AL

(Project No. 4910-0605)

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INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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3.1.1.1 Methods of preservation. Cleaning processes, drying procedures,